

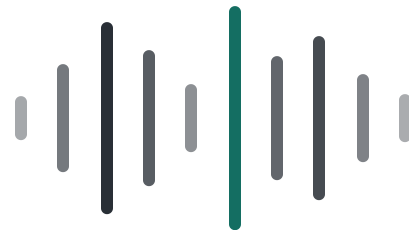
FOR ELEVENLABS AGENTS

Route before the agent speaks.

A computer-vision route signal that helps voice agents answer, verify, or escalate before a single word is spoken.

ElevenLabs gives us the voice loop. ActMap gives that loop a hidden-activation signal for choosing the next path.

INSIDE THE MODEL



BEFORE SPEECH

We read the model's own confidence and turn it into the next action.

THE PROBLEM

The route has to be right before speech.

Voice agents already choose a path before the reply is spoken. The hard part is knowing which path is safe.

ANSWER

Fast when safe

Stable product and navigation questions should speak immediately without tool delay.

VERIFY

Check live facts

Account, billing, policy, and availability questions need RAG, tools, or backend data.

ESCALATE

Hand off early

Fraud, legal, urgent, and angry-customer turns should reach a human before improvisation.

THE INSIGHT

Same words. Different action.

One topic, three correct responses. The right move depends on risk, not the keyword.

ANSWER

"Where is your refund policy?"

Stable public info. Speak instantly, no lookup.

VERIFY

"Am I eligible for a refund?"

Depends on this account. Check the data, then speak.

ESCALATE

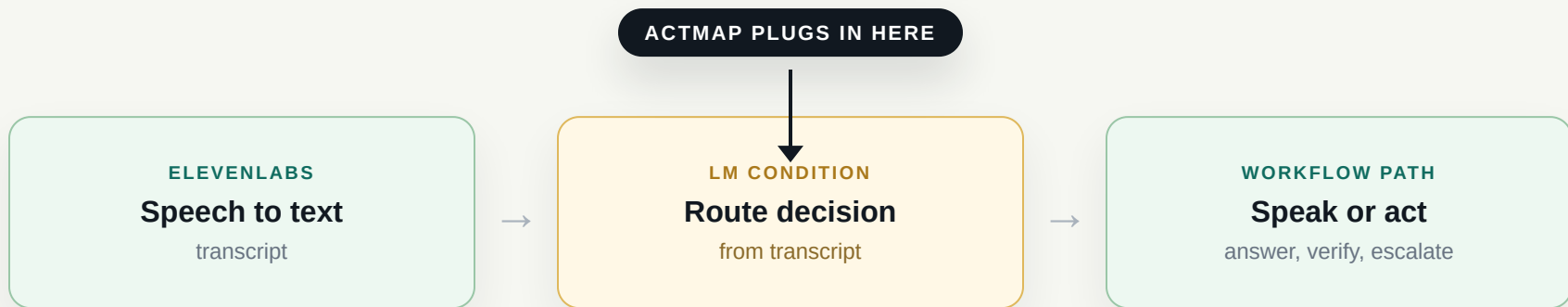
"Someone used my card without permission."

Risk and fraud. Hand to a human, do not improvise.

The topic is similar. The correct route depends on operational risk, not just intent.

ElevenLabs already routes before speech.

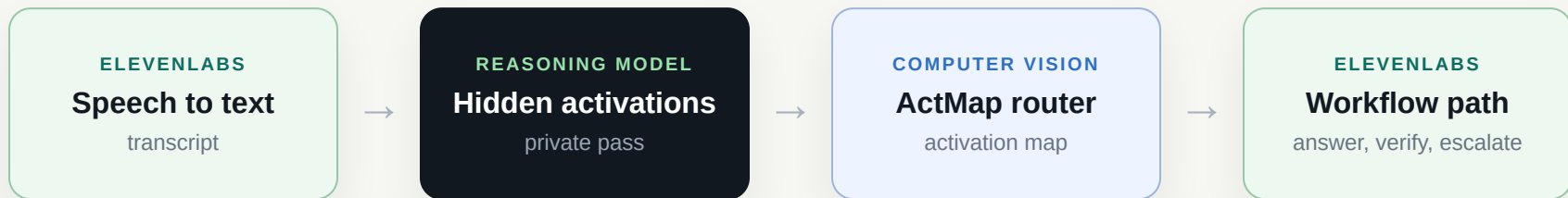
Agent Workflows can choose a path with LLM conditions before text-to-speech runs.



Pre-speech route point: after transcription, before the workflow speaks or acts.

Same route moment. Different model.

ActMap routes from the reasoning model's activations with a computer-vision classifier.



One ElevenLabs loop. Three route outcomes.

Each turn enters through ElevenLabs STT, gets an ActMap route, then returns through the appropriate workflow path.

ANSWER

"Where do I find the conversation simulator?"

Speak now. Stable product guidance goes straight to the voice reply.

VERIFY

"Who mentioned me in comments this month?"

Check first. Tools or RAG fetch live account data before speech.

ESCALATE

"Someone used my card without permission."

Hand off. The caller gets a safe transfer message, not an improvised answer.

THE PROOF

The CV route signal is stronger.

Benchmarked against a Qwen language-model router classifying the same transcripts.

96.2%

ROUTING ACCURACY

On held-out test cases.

+5.2

POINTS MORE ACCURATE

Vs. routing on transcript text.

~250ms

OPTIMIZED ROUTE LATENCY

Warm model, short pass.

3

ROUTES, ONE MODEL

Answer, verify, escalate.

Head to head

ROUTER	ACC.	F1
ActMap CV	96.18	0.962
Transcript LM	91.01	0.911

AT SCALE

Correct routes save real money.

Scenario: 10M voice turns per day, with each avoided retrieval or tool path worth \$0.02 in direct inference cost.

517k

additional correct routes per day

From the measured lift at 10M voice turns/day.

\$10.3k

daily retrieval cost exposure

Corrected routes x \$0.02 per avoided retrieval/tool path.

\$3.8M

annual savings opportunity

Direct retrieval savings only. Avoided support risk is not dollarized.

ActMap: White-Box Uncertainty Quantification from LLM Runtime Activations

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